VEDAVYAS ROKALABOINA

Cardiff, Wales, United Kingdom

+44 7554809910 | vedavyasrokalaboina1@gmail.com | https://www.linkedin.com/in/vedavyas-rokalaboina/ |

https://vedavyasrokalaboina.github.io/

PERSONAL PROFILE

Aeronautical Engineering MSc graduate with strong expertise in simulation, mathematical modelling, and computational analysis, enhanced through aerospace design and UAV development projects. Experienced in CFD, FEA, and performance modelling using ANSYS, MATLAB, and Python, with a proven ability to translate real-world systems into high-fidelity simulation environments. Skilled in analysing complex datasets, validating models, and optimising system performance through numerical methods. Passionate about applying simulation-driven engineering to high-performance environments such as motorsport, with a strong interest in vehicle dynamics, model correlation, and performance optimisation.

SKILLS

Simulation & Analysis: CFD (ANSYS Fluent), FEA, Numerical Modelling, Model Validation

Programming: MATLAB, Python

CAD & Tools: CATIA V5, SolidWorks, XFLR5, Open FOAM

Engineering Areas: Aerodynamics, Flight Mechanics, Performance Analysis

Soft Skills: Problem-solving, Team collaboration, Technical communication

ENGINEERING WORK EXPERIENCE

Technical Lead – UAV Development

Abhuday Bharat Projects Pvt Ltd | Mar 2024 – Sep 2025

- Developed UAV components using SolidWorks and CATIA.
- Performed CFD and FEA analysis using ANSYS Fluent to optimise aerodynamic performance.
- Improved performance models and analysed simulation data using MATLAB.
- Supported prototype development and testing, improving system reliability.
- Collaborated with multidisciplinary teams to deliver engineering solutions.

Technical Lead – Aeronautical Projects (Student Team)

ARDC IARE | Apr 2022 – Feb 2024

- Led aeronautical projects involving CFD, structural analysis, and system modelling.
- Improved simulation models to analyse aerodynamic and flight performance.
- Guided team members in using ANSYS, MATLAB, and CAD tools.
- Delivered technical reports and presentations based on analysis results.

OTHER WORK EXPERIENCE

Customer Care Assistant – Part Time

McDonalds Corporation | APR 2025 – Present

- Delivered high-quality customer service in a fast-paced environment while maintaining efficiency and teamwork.

PROJECTS AND ACCOMPLISHMENTS

SAE INTERNATIONAL AND SAE NATIONAL AERO DESIGN CHALLENGE | JAN 2022 – MAR 2023

- Led CFD, FEA, and CAD activities as Technical Lead for SAE Aero Design projects.
- Accomplished 15th place (International – USA 2023) and 3rd place (National – India 2022).

THRUST VECTOR CONTROL BY JET VANES | JUN 2023 – APR 2024

- Carried out CFD analysis on six different shaped nozzle geometries.
- Analysed 1.8 Mach high-speed flow characteristics and performance.
- Identified Rectangular shaped nozzle as optimal nozzle design based on simulation results.

BUSINESS JET DESIGN, ANALYSIS AND SIMULATION | SEP 2024 – APR 2025

- Designed an 8-passenger aircraft based on mission requirements.
- Estimated take-off weight and key design parameters.
- Used MATLAB for performance calculations and aircraft design modelling.
- Created CAD model and validated performance through SolidWorks simulation and Merlin Simulator.

INVESTIGATION OF THE AERODYNAMICS OF A BIO – INSPIRED BWB | APR 2025 – SEP 2025

- Developed and validated a high-fidelity CFD model of a bio-inspired Blended Wing Body (BWB) aircraft using ANSYS Fluent, including mesh convergence and benchmark validation (<1% error).
- Designed BWB geometry based on established aerodynamic principles, incorporating NACA 23112 and NACA 2412 airfoils and optimised winglet configurations for improved stability.
- Conducted parametric analysis on winglet sweep angles (30°–60°), identifying 38° as optimal, achieving maximum lift-to-drag ratio (L/D).
- Analysed pressure contours and flow behaviour to validate aerodynamic performance, demonstrating efficient lift distribution and minimal flow separation.

EDUCATION

UNIVERSITY OF SOUTH WALES – **Master`s degree | SEP 2024 – SEP 2025**

MSc Aeronautical Engineering.

- Focused on aerodynamics, simulation, propulsion, and computational engineering.
- Applied MATLAB and CFD tools to analyse and optimise engineering systems.

INSTITUTE OF AERONAUTICAL ENGINEERING - Bachelor`s degree | DEC 2020 – APR 2024

Bachelor of Technology Aeronautical Engineering

- Strong grounding in aircraft design, structures, and aerodynamics, simulation tools.